

8.5 Orthogonal Projection



Core skills covered in this section:

- ▶ Constructing a planar projection matrix from the tangent vector.
- ▶ Constructing projection matrices in 2D, 3D, and higher dimensions from the normal vector.
- ▶ The relationship between the tangent vector and the normal vector.
- ▶ The determinant of an orthogonal projection matrix is 0, so it is not invertible.
- ▶ Repeated projection leaves the result unchanged, which shows that an orthogonal projection matrix is idempotent.

Reviewing Vector Projection

As shown in Figure 1, projecting onto a nonzero vector τ (with its starting point at the origin) is equivalent to projecting onto the line through the origin whose tangent vector is τ , and the projection matrix P is

$$P = \frac{\tau @ \tau^T}{\tau^T @ \tau} = \frac{\tau}{\|\tau\|} @ \left(\frac{\tau}{\|\tau\|} \right)^T$$

Intuitively, a line's tangent vector is the arrow that describes "which direction the line runs."

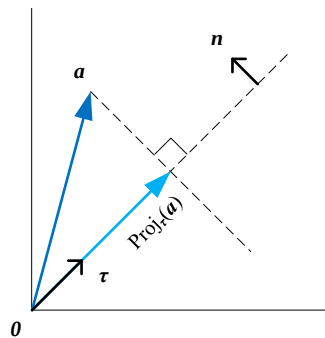


Figure 1. Tangent vector of a line through the origin,

Writing the nonzero vector τ as a direction vector (unit vector), is

$$P = \hat{\tau} @ \hat{\tau}^T$$

The matrix in the equation above is clearly rank one.

As introduced earlier in this book, a rank-one matrix is one that can be expressed as the outer product of a (nonzero) column vector and a (nonzero) row vector; that is, all the rows (or columns) of the matrix are linearly dependent on one another.

⚠ Note that the projection matrix given by is suited to projecting onto a single direction. For orthogonal projection onto a plane/hyperplane in a higher-dimensional space, the normal vector is used to construct the orthogonal projection matrix instead.

This book introduced earlier how to compute vector projection; let's briefly review it below.

As Figure 1 shows, the vector $\mathbf{a}$'s projection onto the vector $\mathbf{\tau}$ is:

$$\text{proj}_{\mathbf{\tau}} \mathbf{a} = \frac{\mathbf{\tau} @ \mathbf{\tau}^T}{\mathbf{\tau}^T @ \mathbf{\tau}} \mathbf{a}$$

Using the vector projection introduced earlier in this book, written in inner-product form, is equivalent to

$$\text{proj}_{\mathbf{\tau}} \mathbf{a} = \underbrace{\|\mathbf{a}\| \cos \theta}_{\text{Scalar projection}} \times \underbrace{\hat{\mathbf{\tau}}}_{\text{Direction vector}} = \frac{\mathbf{a} \cdot \mathbf{\tau}}{\|\mathbf{\tau}\|} \times \frac{\mathbf{\tau}}{\|\mathbf{\tau}\|} = \frac{\mathbf{a} \cdot \mathbf{\tau}}{\|\mathbf{\tau}\|^2} \mathbf{\tau} = \frac{\mathbf{a} \cdot \mathbf{\tau}}{\mathbf{\tau} \cdot \mathbf{\tau}} \mathbf{\tau}$$

A Few Examples

Orthogonally projecting a planar geometric shape onto the horizontal axis ($x_2 = 0$), the tangent vector corresponding to the horizontal axis, $\mathbf{\tau} = [1, 0]^T$, so the projection matrix is

$$\mathbf{P} = \hat{\mathbf{\tau}} @ \hat{\mathbf{\tau}}^T = \begin{bmatrix} 1 \\ 0 \end{bmatrix} @ \begin{bmatrix} 1 & 0 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

The vector $[x_1, x_2]^T$'s orthogonal projection onto the horizontal axis is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$$

This agrees with Figure 2 (a): the x_1 coordinate stays unchanged, and x_2 becomes 0.

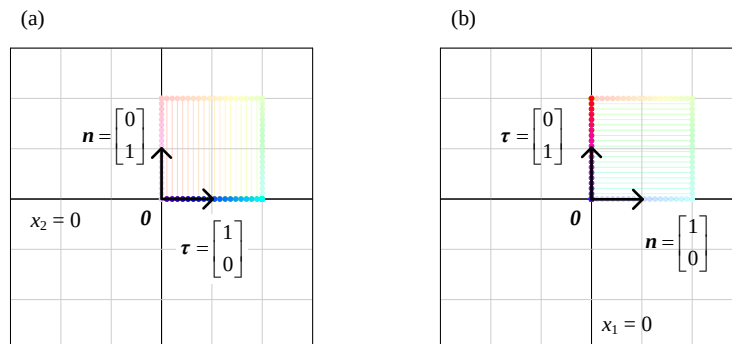


Figure 2. Projecting Onto the Horizontal and Vertical Axes

As Figure 2 (b) shows, orthogonally projecting a planar geometric shape onto the vertical axis ($x_1 = 0$, tangent vector $\tau = [0, 1]^T$) – the projection matrix is

$$P = \hat{\tau} @ \hat{\tau}^T = \begin{bmatrix} 0 \\ 1 \end{bmatrix} @ \begin{bmatrix} 0 \\ 1 \end{bmatrix}^T = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

The vector $[x_1, x_2]^T$'s orthogonal projection onto the vertical axis is

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ x_2 \end{bmatrix}$$

This means the x_1 coordinate becomes 0, while the x_2 coordinate stays unchanged.

Let's look at Figure 3 two more examples.

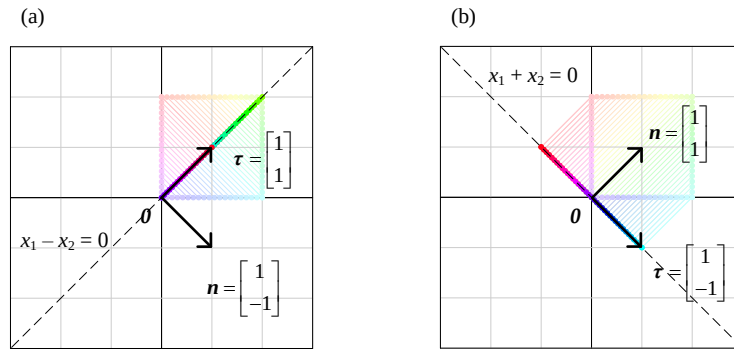


Figure 3. Projecting Onto $x_1 - x_2 = 0$ and $x_1 + x_2 = 0$

As Figure 3(a) shows, orthogonally projecting a planar geometric shape onto $x_1 - x_2 = 0$, the $x_1 - x_2 = 0$ line's corresponding tangent vector $\tau = [1, 1]^T$; normalizing τ gives

$$\hat{\tau} = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}$$

so the projection matrix is

$$P = \hat{\tau} @ \hat{\tau}^T = \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} @ \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix}^T = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$$

The vector $[x_1, x_2]^T$'s orthogonal projection onto $x_1 - x_2 = 0$ changes the coordinates to

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.5x_1 + 0.5x_2 \\ 0.5x_1 + 0.5x_2 \end{bmatrix}$$

The equation above tells us that after projection the horizontal and vertical coordinates are equal, which agrees with Figure 3 (a): all points land on the line $x_1 - x_2 = 0$.

As Figure 3 (b) shows, orthogonally projecting onto $x_1 + x_2 = 0$, the $x_1 + x_2 = 0$ line's corresponding tangent vector $\tau = [1, -1]^T$, so the projection matrix is

$$P = \hat{\tau} @ \hat{\tau}^T = \begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix} @ \begin{bmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{bmatrix}^T = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix}$$

The vector $[x_1, x_2]^T$'s orthogonal projection onto $x_1 + x_2 = 0$ gives coordinates

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.5x_1 - 0.5x_2 \\ -0.5x_1 + 0.5x_2 \end{bmatrix}$$

From the equation above, we see that after projection the horizontal and vertical coordinates are opposites of each other. This agrees with Figure 3 (b): all points land on $x_1 + x_2 = 0$.

Normal Vectors in the Plane

Simply put, a line's normal vector is perpendicular to the line, and a plane's normal vector is perpendicular to the plane.

Given the line $a_1x_1 + a_2x_2 = b$, the corresponding normal vector is

$$\mathbf{n} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$$

Please take some time to analyze on your own the examples given in Figure 4.

⚠ Note that in Figure 4, the lines' normal vectors are all free vectors — their starting points are not at the origin.

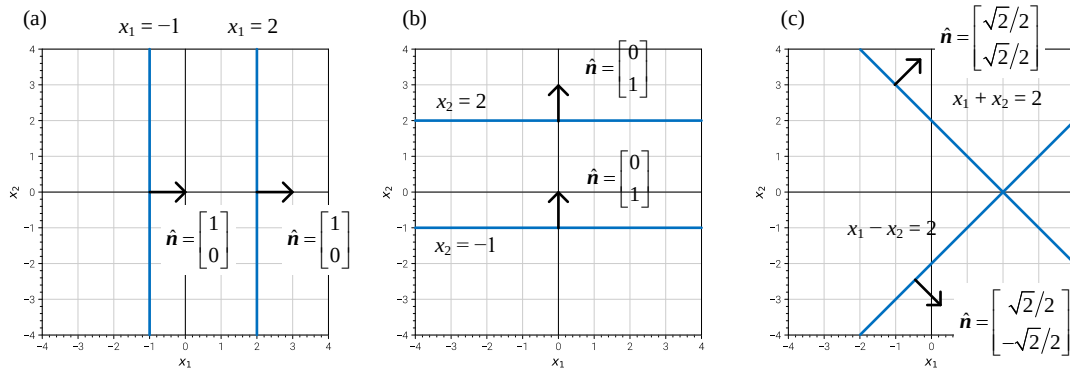


Figure 4. Unit Normal Vectors of Lines in the Plane

Based on orthogonal decomposition of vectors, as Figure 5 shows, we can compute the orthogonal projection of the vector $\mathbf{a}$ onto $\mathbf{n}$ (the normal vector),

$$\text{proj}_{\mathbf{n}} \mathbf{a} = \frac{\mathbf{n} @ \mathbf{n}^T}{\mathbf{n}^T @ \mathbf{n}} \mathbf{a}$$

Then, $\mathbf{a}$ minus the projection $\text{proj}_{\mathbf{n}} \mathbf{a}$, i.e., this vector's projection, is $\mathbf{a}$'s orthogonal projection onto $\boldsymbol{\tau}$.

$$\mathbf{a} - \text{proj}_{\mathbf{n}} \mathbf{a} = \mathbf{a} - \frac{\mathbf{n} @ \mathbf{n}^T}{\mathbf{n}^T @ \mathbf{n}} \mathbf{a} = \left(\mathbf{I} - \frac{\mathbf{n} @ \mathbf{n}^T}{\mathbf{n}^T @ \mathbf{n}} \right) \mathbf{a}$$

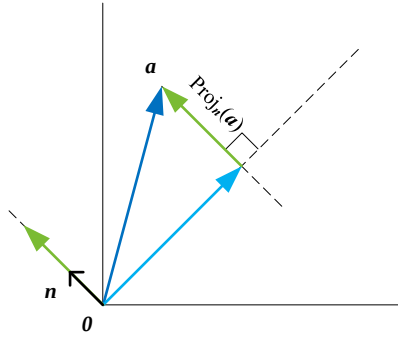


Figure 5. Normal Vector of a Line Through the Origin

So, a 's projection toward n , the projection matrix P is

$$P = I - \frac{n @ n^T}{n^T @ n}$$

⚠ Note that for orthogonal projection in higher-dimensional spaces, we generally use the projection matrix defined via the normal vector, as in ; you will see this shortly, later in this section.

Substituting into gives

$$\begin{aligned} P &= I - \frac{n @ n^T}{n^T @ n} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{1}{a_1^2 + a_2^2} \times \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} @ \begin{bmatrix} a_1 & a_2 \end{bmatrix} \\ &= \frac{1}{a_1^2 + a_2^2} \left(\begin{bmatrix} a_1^2 + a_2^2 & 0 \\ 0 & a_1^2 + a_2^2 \end{bmatrix} - \begin{bmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{bmatrix} \right) \\ &= \frac{1}{a_1^2 + a_2^2} \begin{bmatrix} a_2^2 & -a_1 a_2 \\ -a_1 a_2 & a_1^2 \end{bmatrix} \end{aligned}$$

? Please compute, using both the tangent vector and the normal vector, the projection matrices corresponding to the examples in Figure 6.

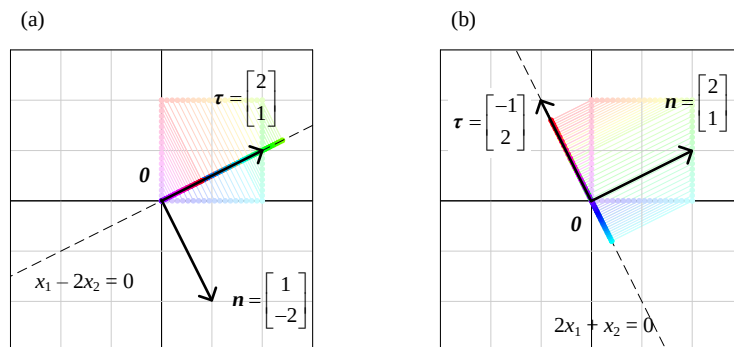


Figure 6. Two Examples of Orthogonal Projection Transformations, with Normal Vector

Let's derive from another angle: if $n = [a_1, a_2]^T$, $\tau = [-a_2, a_1]^T$, using we obtain the same result.

$$\begin{aligned}
 \mathbf{P} &= \frac{\boldsymbol{\tau} @ \boldsymbol{\tau}^T}{\boldsymbol{\tau}^T @ \boldsymbol{\tau}} \\
 &= \frac{\begin{bmatrix} -a_2 \\ a_1 \end{bmatrix} @ \begin{bmatrix} -a_2 & a_1 \end{bmatrix}}{\begin{bmatrix} -a_2 & a_1 \end{bmatrix} @ \begin{bmatrix} -a_2 \\ a_1 \end{bmatrix}} \\
 &= \frac{1}{a_1^2 + a_2^2} \begin{bmatrix} a_2^2 & -a_1 a_2 \\ -a_1 a_2 & a_1^2 \end{bmatrix}
 \end{aligned}$$

Orthogonal Matrix: Orthonormal Basis

Combining and , and simplifying, we get

$$\frac{\mathbf{n} @ \mathbf{n}^T}{\mathbf{n}^T @ \mathbf{n}} + \frac{\boldsymbol{\tau} @ \boldsymbol{\tau}^T}{\boldsymbol{\tau}^T @ \boldsymbol{\tau}} = \mathbf{I}$$

That is,

$$\hat{\mathbf{n}} @ \hat{\mathbf{n}}^T + \hat{\boldsymbol{\tau}} @ \hat{\boldsymbol{\tau}}^T = \mathbf{I}$$

Compute the Gram matrix of the square matrix $\begin{bmatrix} \hat{\mathbf{n}} & \hat{\boldsymbol{\tau}} \end{bmatrix}$.

$$\begin{bmatrix} \hat{\mathbf{n}} & \hat{\boldsymbol{\tau}} \end{bmatrix} @ \begin{bmatrix} \hat{\mathbf{n}} & \hat{\boldsymbol{\tau}} \end{bmatrix}^T = \begin{bmatrix} \hat{\mathbf{n}} & \hat{\boldsymbol{\tau}} \end{bmatrix} @ \begin{bmatrix} \hat{\mathbf{n}}^T \\ \hat{\boldsymbol{\tau}}^T \end{bmatrix} = \hat{\mathbf{n}} @ \hat{\mathbf{n}}^T + \hat{\boldsymbol{\tau}} @ \hat{\boldsymbol{\tau}}^T = \mathbf{I}$$

We find that this Gram matrix is the identity matrix!

This shows that the unit normal vector and tangent vector form an orthonormal basis

$$\begin{bmatrix} \hat{\mathbf{n}} & \hat{\boldsymbol{\tau}} \end{bmatrix}$$

From another perspective, the matrix in the equation above is an orthogonal matrix.

⚠ Note that the equation is very important – it is one of the key points of the next chapter.

Properties of Planar Orthogonal Projection

It is easy to see that the projection matrix is a symmetric matrix

$$\mathbf{P} = \mathbf{P}^T$$

Looking at , we find that the column vectors of the planar orthogonal projection matrix are linearly dependent

$$\mathbf{P} = \frac{1}{a_1^2 + a_2^2} \begin{bmatrix} a_2^2 & -a_1 a_2 \\ -a_1 a_2 & a_1^2 \end{bmatrix} = \begin{bmatrix} a_2 \\ -a_1 \end{bmatrix} \begin{bmatrix} a_2 & -a_1 \end{bmatrix}$$

From we can also see that the planar orthogonal projection matrix $\mathbf{P}$ is rank one, consistent with the conclusion of the equation above.

⚠ Note that $\mathbf{P}$ being rank one applies only to the planar orthogonal projection case; for orthogonal projection in three or more dimensions, the projection matrix is not rank one.

Computing the Determinant of the Planar Orthogonal Projection Matrix

$$\det(\mathbf{P}) = 0$$

As Figure 7 shows, this means that $\mathbf{P}$ compresses the planar geometric shape, causing a loss of information. This also shows that the orthogonal projection matrix is not invertible.

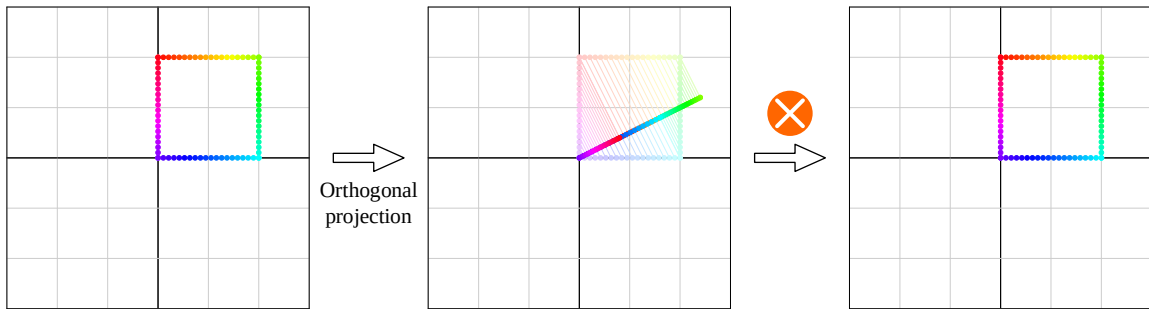


Figure 7. Orthogonal Projection Transformation Is Not Invertible

The projection matrix satisfies

$$\mathbf{P} @ \mathbf{P} = \mathbf{P}$$

We call $\mathbf{P}$ an idempotent matrix; simply put, multiplying an idempotent matrix by itself still gives itself.

As Figure 8 shows, from a geometric point of view, the equation above means that projecting a point multiple times leaves the result unchanged.

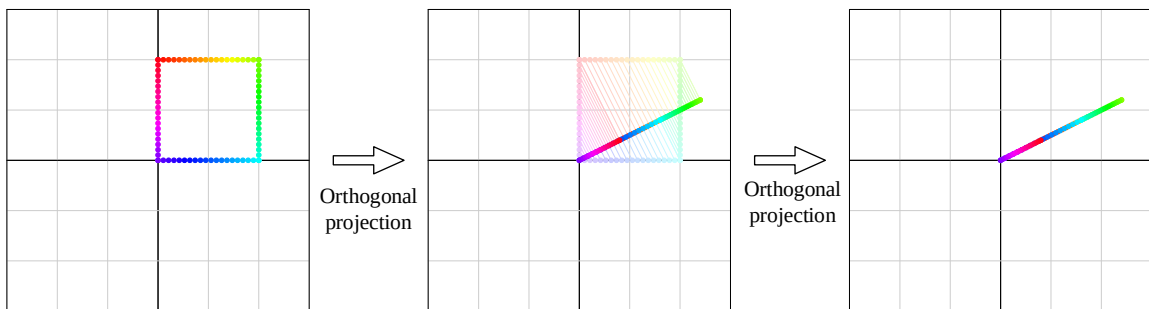


Figure 8. Repeated Orthogonal Projection Leaves the Result Unchanged

Projecting Three-Dimensional Space Onto a Plane

For orthogonal projection in three or more dimensions, the normal vector really shows its advantage!

As Figure 9 shows, for a plane, its normal vector is the vector that is perpendicular to the plane.

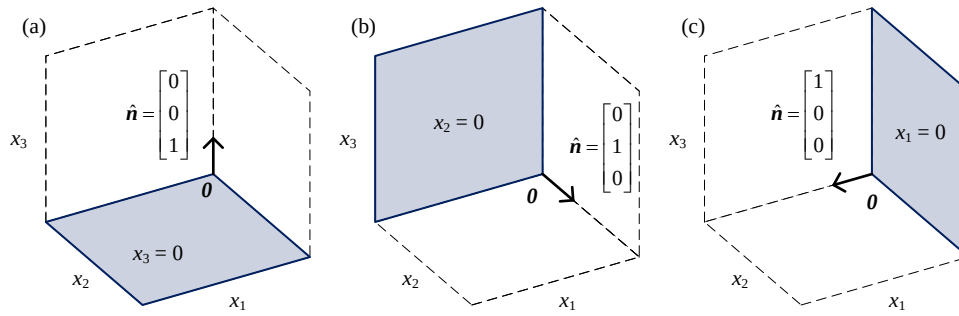


Figure 9. The Normal Vector of a Plane

If a plane's unit normal vector is $\hat{n}$, the orthogonal projection matrix onto this plane, P , is

$$P = I - \hat{n} @ \hat{n}^T$$

As Figure 9 (a) shows, the unit normal vector of the x_1x_2 plane is $[0, 0, 1]^T$, and the orthogonal projection matrix onto the x_1x_2 plane is

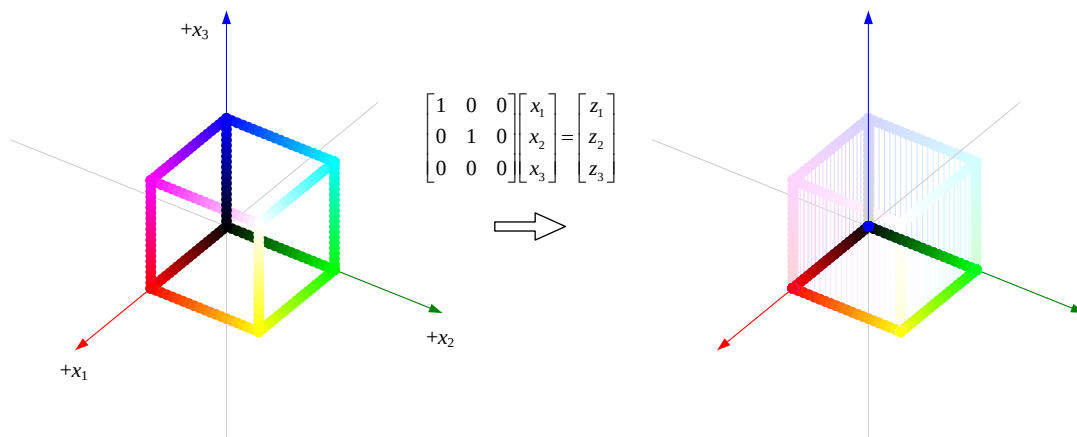
$$P = I - \hat{n} @ \hat{n}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} @ \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

? Please compute the rank and determinant of the orthogonal projection matrix. Is the projection matrix P still rank one?

As Figure 10 shows, the result of orthogonally projecting the three-dimensional column vector x onto the x_1x_2 plane is

$$\begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix}$$

That is, the x_1, x_2 coordinates stay unchanged, and the x_3 coordinate becomes 0.

Figure 10. Orthogonal Projection Onto the x_1x_2 Plane, Three-Dimensional Space

As Figure 9 (b) shows, the unit normal vector of the x_1x_3 plane is $[0, 1, 0]^T$, and the orthogonal projection matrix onto the x_1x_3 plane is

$$\mathbf{P} = \mathbf{I} - \hat{\mathbf{n}}\hat{\mathbf{n}}^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

As Figure 11 shows, the x_1, x_3 coordinates stay unchanged, and the x_2 coordinate becomes 0.

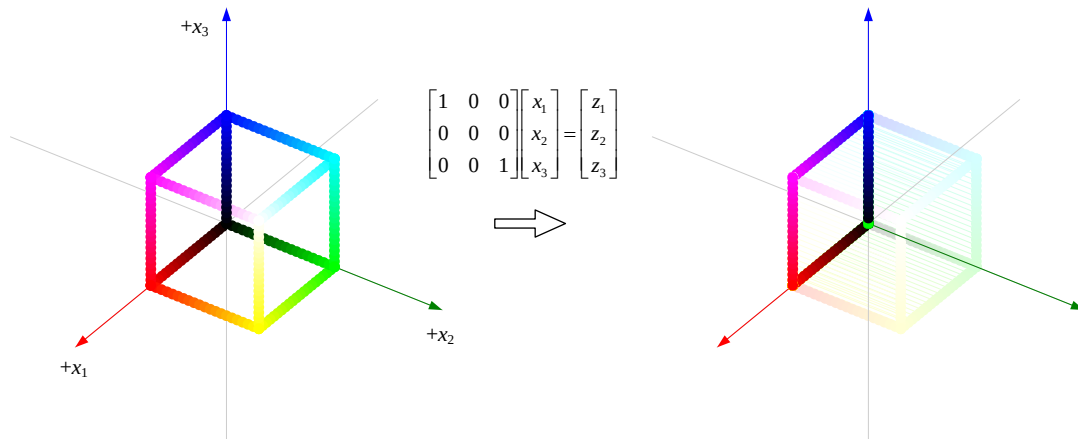


Figure 11. Orthogonal Projection Onto the x_1x_3 Plane, Three-Dimensional Space

? Please compute on your own the orthogonal projection matrix corresponding to Figure 12, and state the resulting coordinate transformation.

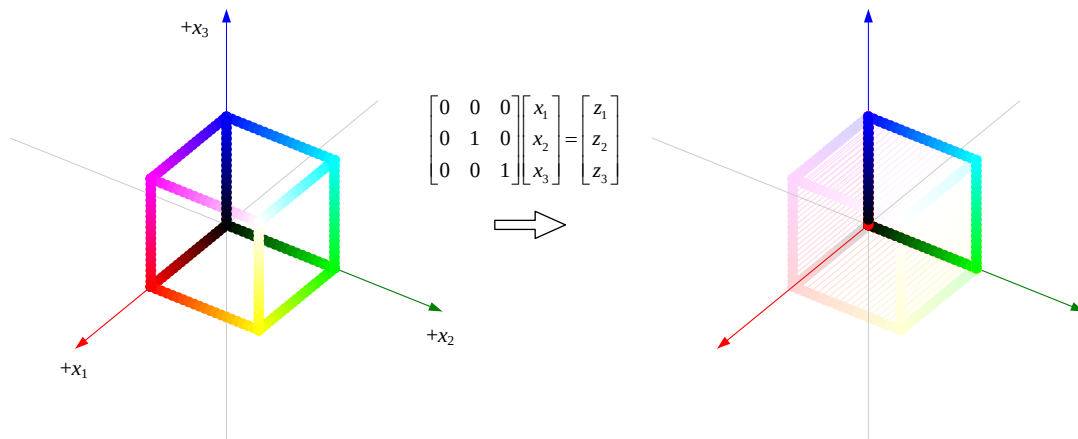


Figure 12. Orthogonal Projection Onto the x_2x_3 Plane, Three-Dimensional Space

You may be curious: where does the information lost in orthogonal projection go? How can this information be recovered? What other use cases does orthogonal projection have? These are all topics explored in the next chapter.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. For each of the following lines through the origin in the plane, compute its tangent vector, normal vector, and projection matrix; please compute the projection matrix using two different methods.

- ▶ $-x_2 = 0$
- ▶ $-x_1 = 0$
- ▶ $x_1 - 2x_2 = 0$
- ▶ $2x_1 - x_2 = 0$
- ▶ $x_1 + 2x_2 = 0$
- ▶ $2x_1 + x_2 = 0$

Q2. For each of the following planes through the origin in three-dimensional space, compute the normal vector and the projection matrix.

- ▶ $x_2 = 0$
- ▶ $x_1 = 0$
- ▶ $x_1 + x_2 = 0$
- ▶ $x_1 + x_3 = 0$
- ▶ $x_2 + x_3 = 0$
- ▶ $x_1 + x_2 + x_3 = 0$
- ▶ $x_1 - x_2 - x_3 = 0$

Q3. Compute the projection of the point (3, 4) onto each of the following unit vectors (i.e., orthogonal projection onto the line through the origin whose tangent vector is the given vector):

- ▶ $[0, 1]^T$
- ▶ $[1, 0]^T$
- ▶ $[3/5, 4/5]^T$
- ▶ $[4/5, -3/5]^T$

Q4. A line through the origin makes a 30-degree angle with the positive horizontal axis. Compute the orthogonal projection of the point (3, 4) onto this line.

Q5. Please modify LA_08_02_01.ipynb to visualize the orthogonal projection operations in Figure 2 and Figure 3.

Q6. Please modify LA_08_03_01.ipynb to visualize the three-dimensional orthogonal projection operations in Figure 10, Figure 11, and Figure 12.

Q7. For each of the following hyperplane normal vectors, compute the projection matrix.

- ▶ $[0, 0, 0, 1]^T$
- ▶ $[0, 0, 1, 0]^T$
- ▶ $[1, 1, 1, 1]^T$

Q8. For each of the following hyperplanes, compute the projection matrix.

► $x_1 + x_2 + x_3 + x_4 = 0$

► $x_1 - x_2 - x_3 + x_4 = 0$

Q9. For projecting onto a line in the plane that does not pass through the origin, think about how to compute the projected point.